09 — Asset Dimensions & Re-Export Targets

Single source of truth for **pixel dimensions, aspect ratios, and re-export rules** for every raw image asset that the Flame game loads. Read this before cropping, re-exporting, or swapping any sprite — the rendering code is calibrated to specific proportions and will misalign if the numbers drift.

Scope: **all sprite-rendered assets.** Board sprites now exist for sets 1–3; checkers and dice remain set-1-only (sets 2–3 fall back to procedural rendering for those).

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1. Current asset inventory

Asset	File	Pixels	Ratio	Notes
Board (set 1 — mahogany & olive)	`set1_board.png`	**2102 × 1950**	1.0779	Re-exported 2026-04-19 from shared 4320×2700 template
Board (set 2 — walnut & navy)	`set2_board.png`	**2102 × 1950**	1.0779	Pixel-aligned with set 1 (same template, recolored triangles/felt)
Board (set 3 — teal & maple)	`set3_board.png`	**2102 × 1950**	1.0779	Pixel-aligned with set 1 (same template, recolored triangles/felt)
Board shadow (set 1)	`set1_board_shadow.png`	2106 × 1954	1.0778	Legacy, 4 px wider than new boards (negligible)
Table backdrop	`backgrounds/table.png`	**4320 × 2700**	1.60 (8:5)	Wood-table texture, ready to wire as game-screen backdrop
Light checker (normal)	`set1_light_normal.png`	293 × 293	1.00 (square)	
Dark checker (normal)	`set1_dark_normal.png`	294 × 294	1.00 (square)	
Light checker (selected)	`set1_light_selected.png`	**369 × 389**	0.949 (NOT square)	Misalignment hotspot
Dark checker (selected)	`set1_dark_selected.png`	**369 × 389**	0.949 (NOT square)	Misalignment hotspot
Die face 1–6	`set1_die_N_active.png`	135–136 × 135	~1.00	1 px drift between faces

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1a. Board template (all 3 sets share this)

The three board sprites are derived from a single shared **4320 × 2700 master PNG** where the board art occupies a fixed bounding box `(1103, 379) → (3205, 2329)` = **2102 × 1950 px**. All three exports (`mahogany_olive`, `walnut_navy`, `teal_maple`) were produced from this template with only the triangle and felt colors varied; the frame, hinge, gold clasps, bar, and triangle geometry are pixel-identical.

> **Verified alignment after cropping to the board bbox:**
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> - Top-frame mean pixel diff across all 3 boards: **7 RGB units** (essentially identical)
> - Structural frame-column mask: **1–2 columns out of 2102 differ** (0.05–0.10%)
> - Gold clasp y-positions: **identical on all 3 boards**

Implication: checkers rendered at any point on any board land in the same screen position. Switching themes mid-game produces no visual drift. The percentages in `board_layout.dart` calibrated against the original 2106 × 1954 sprite apply unchanged to the new 2102 × 1950 sprites (ratio difference is 0.0001, inside rounding noise).

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2. Why misalignment happens

The Flame layout math in `board_layout.dart:53–77` hard-codes percentages measured **directly against the shared board template at ~2100 × 1950**:

Parameter	Current value	Measured from sprite
Frame inner edge	3.1% of board width	65 / 2106 px
Center bar	3.1% of board width	66 / 2106 px
Top/bottom felt inset	2.8% of board height	55 / 1954 px
Triangle row height	32% of board height	625 / 1954 px
Stack anchor nudge	2.0 px	Triangle base shadow depth

> **If a board sprite is re-exported with different proportions, checkers drift off the painted triangles.** Any crop must preserve these internal percentages.

Two additional hotspots:

- **Selected checker sprites are 369 × 389 (not square).** Flame renders them into a square box (`radius × 2`) at `checker_component.dart:185`, which stretches the art and shifts its visual center off the point's anchor. This is the most likely cause of the select-state misalignment.
- **Board shadow is offset by hard-coded `+2, +3 px`** at `board_component.dart:110–117`. Shadow file stays pixel-close to the board file.

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3. Re-export targets

3.1 Board + board shadow

- **Ratio:** 1.0779 : 1 (current)
- **Size: 2102 × 1950 px** (cropped from the shared 4320 × 2700 master)
- **Internal proportions preserved from the shared template** — percentages in `board_layout.dart` remain valid:
  - Frame inner edge: ~3.1% of board width
  - Center bar: ~3.1% of board width
  - Top/bottom felt inset: ~2.8% of board height
  - Triangle row height: ~32% of board height
  - Triangle base shadow: ~2 px
- **All new sets must come from the same 4320 × 2700 master** cropped to the same bbox `(1103, 379) → (3205, 2329)`. This guarantees pixel-alignment across themes.
- Shadow (`set1_board_shadow.png`) is still 2106 × 1954 from the earlier master — the 4 px discrepancy is visually negligible but can be re-cropped to match later.

3.2 Checkers (all 4 variants)

- **Ratio:** 1:1 square — non-negotiable
- **Size: 400 × 400 px** for every variant (normal + selected, light + dark)
- Chip artwork must be **centered** with **identical padding** across all 4 files
- Selected-state glow/halo must fit **inside** the same 400 × 400 canvas — do not grow the canvas
- This single change (selected variants becoming 1:1 instead of 0.949:1) resolves the select-state jump

3.3 Dice (6 faces)

- **Ratio:** 1:1 square
- **Size: 400 × 400 px** (consistent with checkers — one canvas size for all sprite assets)
- All 6 faces must be **exactly** the same dimensions (no 1 px drift)
- Pips must sit in the **same safe area on every face** so the die doesn't jitter when the value changes

3.4 Background (table.png now available)

`table.png` (4320 × 2700, 1.60 ratio) is now in `assets/images/backgrounds/` — a warm wood texture ready to be wired as the game-screen backdrop. The game still renders a flat color fill at `game_screen.dart:594`; switching to `table.png` is a separate task.

General rules for any background image:

- **Ratio: 21:9 (2.33:1)** — matches the widest phone so nothing important gets cropped off the ends
- **Size: 2520 × 1080 px**
- `BoxFit.cover` — image scales to fill the landscape viewport

Safe zones (content that must survive cropping on narrower devices):

Target	Safe zone	Devices covered
Full canvas	2520 × 1080 (21:9)	Ultrawide Android phones
16:9 safe zone	center 1920 × 1080	Standard 16:9 phones
4:3 safe zone	center 1440 × 1080	iPad / tablets landscape

Board occlusion zone: the board covers the center ~2420 × 840 of the viewport. Nothing inside that zone is ever visible — keep it as a neutral texture and put visual interest at the **edges** and **top/bottom bands**.

Color direction: current fill is `#6B4F3A` (warm brown). Replacement imagery should sit in that warm-earth family so the UI text and buttons (designed against this color) keep contrast per `00_TOKENS.md`.

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4. Summary rules

Asset	Ratio	Size	Rule
Board (all sets)	1.0779:1	2102 × 1950	Must come from the shared 4320×2700 master cropped to bbox `(1103,379)→(3205,2329)` so all sets align pixel-for-pixel
Board shadow	1.0778:1	2106 × 1954	Set-1-only; reuse for all sets or re-crop to 2102×1950 later
All checkers	1:1	400 × 400	Centered artwork, identical padding, selection glow inside the canvas
All dice	1:1	400 × 400	Identical dimensions across all 6 faces, pips in a shared safe area
Background	1.60:1 (or 21:9)	table.png 4320×2700, or re-export 2520×1080	Visual interest at edges; center ~80% is covered by the board

Hold these rules (square checkers, square dice, template-cropped boards) and the existing rendering math keeps working without code changes. Overlays layer cleanly on top.

5. Files to re-measure if proportions change

If any re-export changes internal proportions (not just resolution), re-measure and update:

- `board_layout.dart:68–77` — frame/bar/inset/triangle percentages
- `board_layout.dart:77` — `stackAnchorNudge` if triangle base shadow changes
- `board_component.dart:110–117` — shadow `+2, +3 px` offset if shadow art changes
- `checker_component.dart:185` — sprite render size math, if checker canvas becomes non-square